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## **Experiment 2**

### **PAAS in Public Cloud Using Render**

#### **AIM**

To build a basic Python Flask web application and deploy it on the Render cloud platform. This experiment helps in understanding how a Platform as a Service (PaaS) allows applications to run online without manually setting up or maintaining servers.

#### **What is PaaS?**

Platform as a Service (PaaS) is a cloud service that provides everything needed to develop, deploy, and run an application. It takes care of server management, software installation, and deployment, allowing developers to focus only on writing the application.

#### **Why Render?**

Render is an easy-to-use cloud hosting platform that supports web application deployment directly from GitHub. It automatically builds and hosts the application, making deployment simple even for beginners.

#### **PROCEDURE:**

##### **Step 1**

Sign in to your GitHub account or create one if you do not already have an account.

##### **Step 2**

Create a project folder named **RenderHelloApp**.

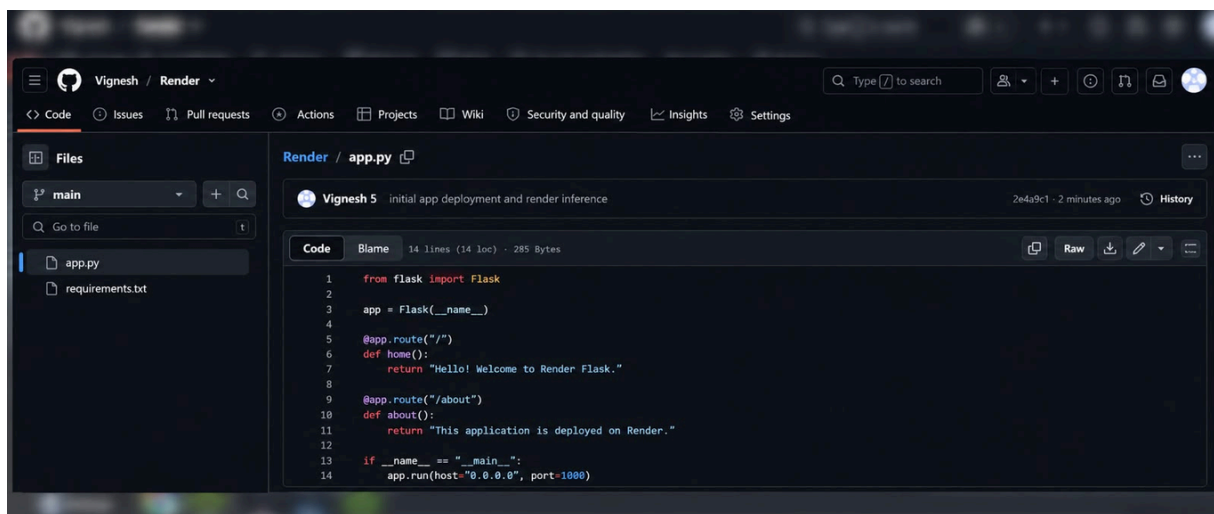
### Step 3

Inside the project folder, create the following files:

- app.py
- requirements.txt

### Step 4

Write the Flask application code in the app.py file.

A screenshot of a GitHub repository interface. The left sidebar shows the file explorer with 'main' selected and files 'app.py' and 'requirements.txt'. The main area displays the 'app.py' file with 14 lines of Python code. The code imports Flask, creates an app, and defines two routes: a home page and an about page. It also includes a main block to run the app on port 1000.

```
1 from flask import Flask
2
3 app = Flask(__name__)
4
5 @app.route("/")
6 def home():
7     return "Hello! Welcome to Render Flask."
8
9 @app.route("/about")
10 def about():
11     return "This application is deployed on Render."
12
13 if __name__ == "__main__":
14     app.run(host="0.0.0.0", port=1000)
```

### Step 5

Add the required Python packages such as **Flask** and **Gunicorn** to the requirements.txt file.

### Step 6

Create a new public repository called **RenderHelloApp** on GitHub.

### Step 7

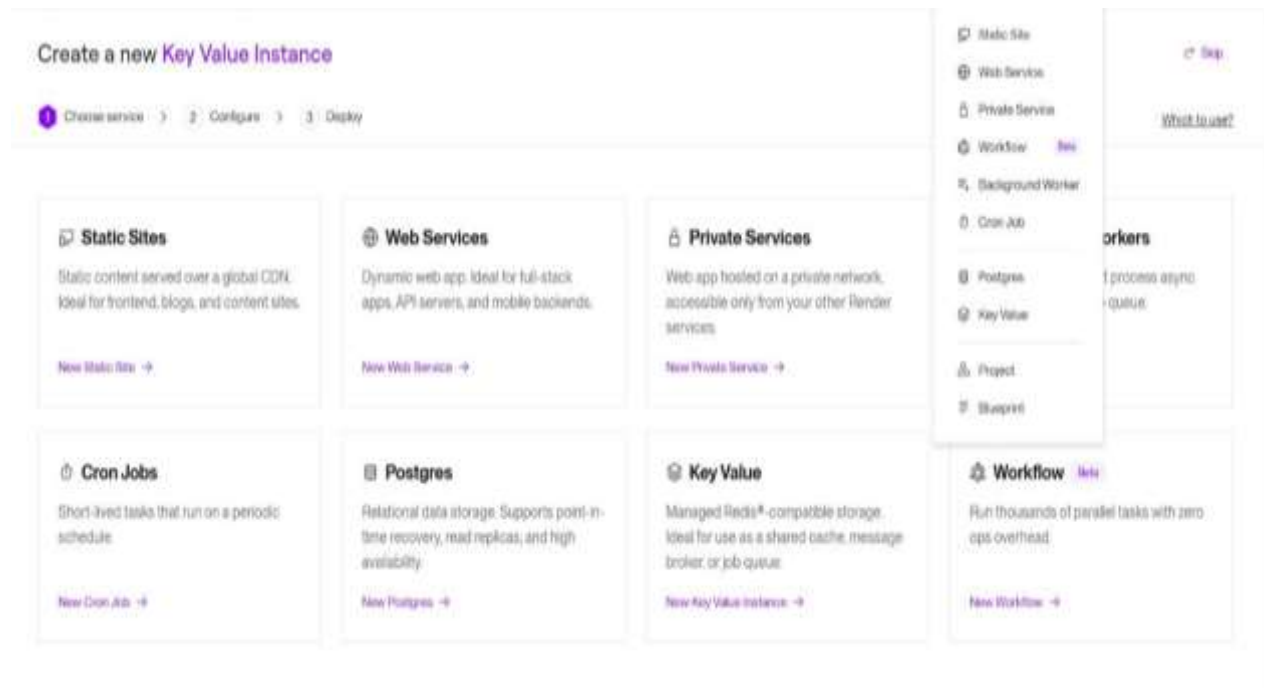
Upload both project files to the repository and commit the changes.

### Step 8

Create an account on Render and connect it to your GitHub account.

## Step 9

Create a new Web Service in Render.



## Step 10

Select the **RenderHelloApp** repository from GitHub.

## Step 11

Enter the deployment details:

- **Language:** Python
- **Branch:** main
- **Root Directory:** Leave empty
- **Build Command:** pip install -r requirements.txt
- **Start Command:** gunicorn app:app
- **Instance Type:** Free (if available)

**Root Directory** Optional  
 If set, Render runs commands from this directory instead of the repository root. Additionally, code changes outside of this directory do not trigger an auto-deploy. Most commonly used with a [monorepo](#).

**Build Command**  
 Render runs this command to build your app before each deploy.

**Start Command**  
 Render runs this command to start your app with each deploy.

**Instance Type**

|                                                                                                                                                                                                                                     |                                                          |                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>For hobby projects</b><br><br>For more power and to get the most out of Render, we recommend using one of our paid instance types. All paid instances support: <ul style="list-style-type: none"> <li>• Zero Downtime</li> </ul> | <b>Free</b><br>\$0 / month<br>512 MB (RAM)<br>0.1 CPU    | <b>Upgrade to enable more features</b><br><small>Free instances spin down after periods of inactivity. They do not support SSH access, scaling, one-off jobs, or persistent disks. Select any paid instance type to enable these features.</small> |
|                                                                                                                                                                                                                                     | <b>Starter</b><br>\$7 / month<br>512 MB (RAM)<br>0.5 CPU |                                                                                                                                                                                                                                                    |
|                                                                                                                                                                                                                                     | <b>Pro</b><br>4 GB (RAM)                                 | <b>Pro Plus</b><br>8 GB (RAM)                                                                                                                                                                                                                      |

## Step 12

Click Deploy Web Service and wait for the deployment to complete.

**Environment Variables**  
 Set environment-specific config and secrets (such as API keys), then read those values from your code. [Learn more](#).

Generate

+ Add Environment Variable
+ Add from env

> Advanced

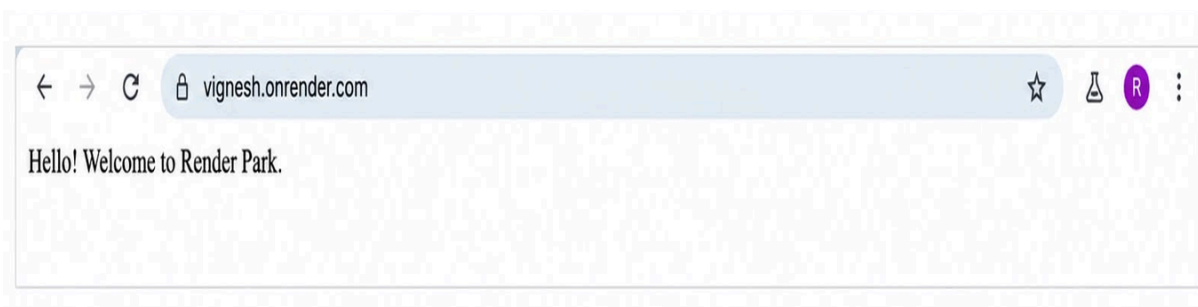
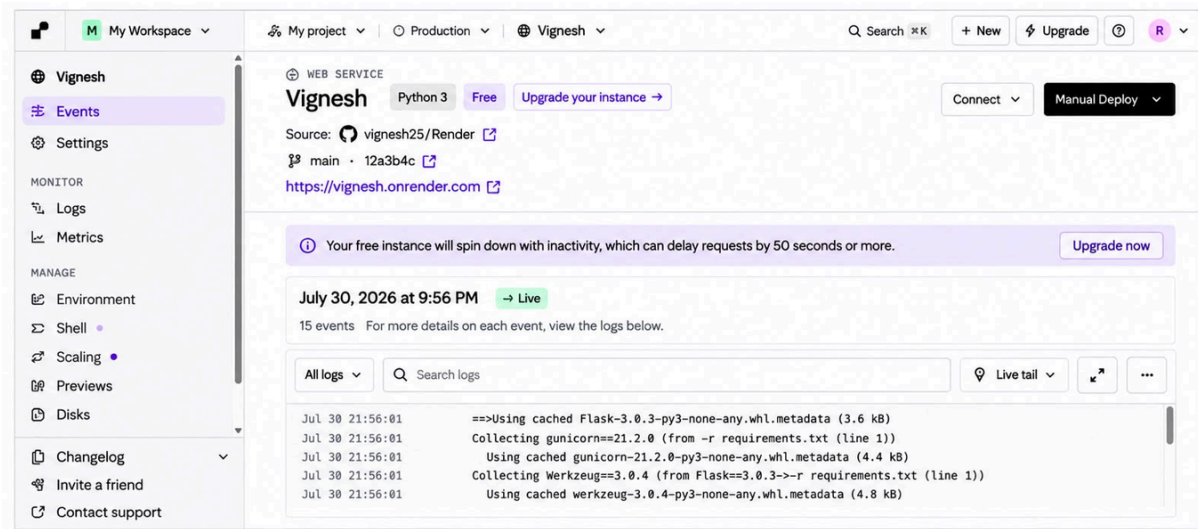
Deploy Web Service

## Step 13

After deployment, open the URL generated by Render and check whether the home page is displayed correctly.

## Step 14

Add **/about** at the end of the application URL to verify that the second route is also working properly.



## CONCLUSION:

This experiment demonstrated how a Flask application can be deployed using Render, a Platform as a Service. I learned that PaaS simplifies application deployment by handling the server setup and hosting, allowing developers to publish applications quickly without managing infrastructure.